NorthByNorthwest

NXNW — Continuum of Care Master Data Engine for the Rogue River Valley

Rogue Raise #02: The Unhoused · Ashland, OR · Oct 23–25, 2026
Version 1.0 · Python 3.10+ · Offline-First · HUD & Non-HUD Sources

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1. Purpose

The annual HUD Point-in-Time (PIT) homeless count produces a single-night snapshot in January that systematically undercounts the true scale of homelessness in the Rogue River Valley (RRV). Agencies funded by HUD operate in rigid HMIS silos, while community-based providers — syringe exchanges, meal programs, recovery cafes, faith-based outreach — capture service activity in disconnected spreadsheets or paper logs that never reach the formal count.

NXNW is an offline-first, pure-Python record linkage engine that:

- De-duplicates client records across HUD and non-HUD data sources using probabilistic matching (Splink + DuckDB).
- Assigns each unique individual a persistent, deterministic Master Person ID (UUID v5) — the foundation of a Rogue River Valley Master Person Index (MPI).
- Aggregates continuous service encounters into a unified fact table spanning shelter, food, harm-reduction, and recovery services.
- Exposes a 'Tinder-style' human-review interface (app.py) so case managers and data stewards can accept or reject proposed identity merges with a single click, maintaining a complete audit trail.
- Exports a county-ready GOLDEN_PERSONS_ENCOUNTERS.csv combining verified persons with every service encounter — suitable for public health dashboards, commissioner briefings, and HUD Consolidated Plan submissions.

Target audience: Jackson County CoC staff, ACCESS case managers, Rogue Retreat intake coordinators, Recovery Café leadership, JCPH harm-reduction program officers, and county commissioners seeking unduplicated population counts.

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2. Installation

Requirements

- Python 3.10 or higher
- pip (standard with Python)
- Internet access for initial package download only; all pipeline execution is fully offline thereafter

Install dependencies

```
pip install streamlit splink duckdb networkx pandas rapidfuzz reportlab
```

Generate synthetic test data (first run only)

```
python generate_access_data.py
python generate_rogue_retreat.py
python generate_recovery_cafe.py
python generate_syringe_exchange.py
python generate_stvdp.py
```

Run the data pipeline

```
python mpi_pipeline.py # full run (stages 1–5)
python mpi_pipeline.py --from-stage 3 # resume from matching
```

Launch the review UI

```
streamlit run app.py
```

Open http://localhost:8501 in a browser. The sidebar shows ingested datasets and live session metrics. Upload a partner CSV to begin the swipe-deck entity-resolution workflow.

Generate the Golden Export (optional)

```
python generate_golden.py
```

Or use the ■ **EXPORT** → **Overwrite Golden File** button in the NXNW sidebar after the pipeline has produced its three output CSVs.

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3. Default Use Case

The canonical workflow proceeds in five stages, each validated by automated agentic checks before the next begins.

Stage 1 — Ingest

Drop provider CSVs into the input/ directory. ingest.py normalises column names across all four agency schemas and produces a unified demographic frame plus a raw service-encounters frame.

Stage 2 — Clean

clean.py strips punctuation from names, standardises all dates to YYYY-MM-DD, extracts SSN last-4, and lower-cases all text fields for comparison.

Stage 3 — Match

match.py runs Splink probabilistic linkage with Levenshtein comparisons on first name, last name, exact DOB, and exact SSN last-4. Blocking rules constrain the candidate pair space. Match probabilities below 0.70 are discarded automatically.

Stage 4 — Cluster

cluster.py feeds accepted match pairs into a NetworkX connected-components graph. Each cluster receives a deterministic UUID v5 Master Person ID seeded from the sorted member record IDs — the same data always produces the same MPI.

Stage 5 — Export

export.py writes dim_master_persons.csv (one row per unique person), bridge_record_links.csv (audit trace), and fact_encounters.csv (all service activity linked to master IDs). Run generate_golden.py or click the sidebar Export button to produce the combined GOLDEN_PERSONS_ENCOUNTERS.csv.

Human review loop: Upload any partner CSV in the NXNW UI. The engine proposes candidate pairs above the minimum confidence threshold. A data steward swipes right (Accept) to merge identities or left (Reject) to sever the link. Every decision is logged in the in-session audit trail and immediately reflected in the live Master Person count.

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4. File Naming Convention

All provider data files follow a two-part snake_case convention: **<agency_slug>_<file_type>.csv**. The agency slug must be unique across all ingested sources; the file type is either **clients** (demographic record) or **services** (encounter/activity log).

Agency / Program	Demographic File	Services File
ACCESS HMIS (HUD)	access_hmis_client.csv	access_hmis_services.csv
Rogue Retreat	rogue_retreat_clients.csv	rogue_retreat_services.csv
Recovery Café	recovery_cafe_members.csv	recovery_cafe_services.csv
Jackson Co. Syringe Exch.	jackson_county_ssp_log.csv	jackson_county_ssp_services.csv
St. Vincent de Paul	stvdp_clients.csv	stvdp_services.csv
New agency template	<agency_slug>_clients.csv	<agency_slug>_services.csv

- Use only lowercase letters, digits, and underscores in slugs. No spaces, hyphens, or special characters.
- Keep slugs consistent between the client file and its matching services file — the pipeline joins them on slug pattern.
- Pipeline output files are always written to the **output/** directory. Do not place provider source CSVs in output/.
- The golden export GOLDEN_PERSONS_ENCOUNTERS.csv always writes to the project root. This is the file for external distribution.

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5. CSV Header Layouts

Each provider must supply a demographics file and, optionally, a services file. Headers below are the canonical names NXNW expects after column mapping. The Column Mapper UI in NXNW lets operators remap non-standard headers at upload time without modifying source files.

5a. Demographic / Client File (all providers)

Column	Required?	Description
source_id	Required	Unique client/member ID within the source agency
first_name	Required	Legal or preferred first name
last_name	Required	Legal or preferred surname
date_of_birth	Required	ISO 8601 preferred (YYYY-MM-DD); common variants parsed
ssn_last4	Optional	Last 4 digits of SSN — strongest linkage signal
phone	Optional	Any format accepted; digits extracted for matching
intake_date	Optional	First service contact date at this agency
gender	Optional	Self-reported; not used in matching
race_ethnicity	Optional	Self-reported; not used in matching
veteran_status	Optional	Y/N; not used in matching

Column	Required?	Description
notes	Optional	Free text; stripped before export

5b. Services / Encounters File (all providers)

Column	Required?	Description
encounter_id	Required	Unique ID for this service event (visit, log, entry)
source_id	Required	FK → demographic file client/member ID
encounter_date	Required	Date service was provided (YYYY-MM-DD preferred)
service_category	Required	Standardised service label (see taxonomy below)
service_type	Optional	More granular sub-type within category
location_info	Optional	ZIP, address, or site name
quantity	Optional	Meals served, syringes exchanged, lbs food, etc.
exit_date	Optional	Shelter/program exit date (residential providers)
case_manager	Optional	Staff name or ID; stripped before export

5c. Pipeline Output Files

Output File	Headers
dim_master_persons.csv	master_person_id, canonical_first_name, canonical_last_name, canonical_dob, canonical_ssn_last4, total_linked_records, linked_sources_count, contributing_sources
bridge_record_links.csv	unique_record_id, master_person_id, source_system, source_id, match_confidence, match_type
fact_encounters.csv	encounter_id, master_person_id, unique_record_id, source_system, encounter_date, service_category, location_info
GOLDEN_PERSONS_ENCOUNTERS.csv	master_person_id, canonical_first_name, canonical_last_name, canonical_dob, contributing_sources, total_linked_records, linked_sources_count, encounter_id, source_system, encounter_date, service_category, location_info

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6. System Limitations

6a. Scale Estimate — Rogue River Valley

The RRV encompasses Medford, Ashland, Grants Pass, Jacksonville, and surrounding communities with a combined population of approximately 200,000. Oregon consistently reports one of the highest per-capita homelessness rates in the United States (~0.28% PIT, per 2023 HUD AHAR). Applying that rate:

Metric	Estimate	Basis
PIT Count (single night, Jan)	~560	200,000 × 0.28%
Annual shelter contacts	~1,400–2,000	2.5–3.5× PIT (HUD shelter utilisation ratio)
Non-HUD service contacts	~1,500–3,000	Meal programs, SSP, warming centers, street outreach

Metric	Estimate	Basis
Est. unique individuals/year	`~2,500–4,000`	`Deduplicated across all programs; NXNW target range`
Service encounter rows/year	`~15,000–40,000`	`Avg 6–10 contacts per active individual`
Projected system record count	`~5,000–8,000`	`Accumulated over 2–3 year rolling window`

At 5,000–8,000 demographic records, Splink with DuckDB backend and the three configured blocking rules produces roughly 80,000–200,000 candidate pairs — well within DuckDB's in-process capacity on a standard laptop with no network dependency. Full pipeline runtime at this scale is estimated at 2–8 minutes.

6b. Record Linkage Accuracy

- Probabilistic matching is not deterministic truth. The default 70% match probability threshold will produce both false positives (two different people merged) and false negatives (the same person left as two records).
- Name-only matches are high-risk. When SSN last-4 and DOB are both absent, first/last name similarity alone generates a high false-positive rate for common names (e.g., Maria Garcia, James Brown).
- Nickname and alias handling is limited. 'Bob' vs 'Robert', 'Liz' vs 'Elizabeth' will not match unless the provider has normalised preferred names.
- Typos and transcription errors in DOB (day/month transposition, wrong year) will suppress correct matches. Fallback blocking on last name compensates partially but increases candidate pair volume.

6c. Anonymous & Low-Identity Records

- Syringe exchange programs (SSP) and many meal/warming programs collect only initials and birth year for harm-reduction safety. These records cannot be probabilistically matched to full-identity records — they are aggregated as anonymous encounters only.
- Records with no shared field values (no name, no DOB, no SSN, no phone) are isolated nodes in the identity graph and will always appear as single-record MPI entries.

■ *Do not use anonymous SSP or warming-shelter records for individual-level case management — they are valid only for population-level aggregate counts.*

6d. Ingestion Constraints

- File size: the Streamlit uploader is capped at 200 MB per file. Providers with large multi-year extracts should split by fiscal year.
- Encoding: UTF-8 is expected. Windows-1252 / Latin-1 files with special characters (accented names) may produce garbled output. Convert with 'iconv' or Excel Save-As UTF-8 CSV before upload.
- Date formats: the cleaner accepts ISO 8601, MM/DD/YYYY, and M/D/YY. Ambiguous two-digit years (e.g., 01/02/03) are interpreted as 2003-02-01 (ISO order). Verify edge cases manually.
- Column mapping: the UI Column Mapper handles up to 9 semantic field slots. Providers with more than 9 meaningful fields must pre-process to the canonical 9-column demographic schema before ingestion.
- Services files are optional. If a provider has no structured service log, the intake_date in the demographics file is used as a single proxy encounter date.

6e. Privacy & Compliance

- NXNW is designed for use by authorised CoC staff and data stewards only. The output CSVs contain PII (name, DOB, SSN last-4). Restrict file system access accordingly.

- The system does not currently implement field-level encryption at rest. Enable the SHA-256 hashing toggle in the UI before sharing any export outside of authorised staff.
- HMIS data shared under a signed Data Sharing Agreement (DSA) must remain within the DSA's permitted use scope. Do not upload HMIS data to any cloud service without explicit DSA authorisation.

■ *NXNW is a decision-support tool, not a legal identity determination system. All merge decisions affecting housing-voucher eligibility must be reviewed by a qualified case manager.*

6f. Infrastructure & Offline Constraints

- All computation is local. No data leaves the machine running NXNW.
- Concurrent multi-user access is not supported. Streamlit serves a single-session process; simultaneous users will share session state unpredictably.
- The identity graph (NetworkX) is rebuilt in memory on each pipeline run. There is no incremental / delta-update mode in the current version — adding a new provider requires a full re-run from Stage 1.
- Browser session state (swipe decisions, threshold settings) is lost on page refresh. Always export before closing the browser tab.